

Streaming Subscriber Churn Prediction & Next-Best-Action Engine

Decision Analytics | Predictive modeling, behavioral segmentation, and ROI-based retention recommendations for a subscription streaming platform.

5,000 Subscribers analyzed	0.88 Model accuracy (ROC-AUC)	2,874 At-risk subscribers identified	~\$85K Estimated annual savings
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Overview

Subscription streaming platforms typically address customer churn with a single, undifferentiated retention offer applied uniformly across all at-risk users. This project develops a more targeted approach: a predictive model identifies which subscribers are likely to cancel, a segmentation layer groups at-risk subscribers by the underlying behavioral driver of that risk, and a decision framework assigns each segment a specific, cost-appropriate retention action with an associated revenue-impact estimate.

Methodology

- Data preparation: cleaned and structured a 5,000-subscriber dataset covering watch time, days since last login, subscription tier, monthly fee, and account tenure indicators.
- Predictive modeling: trained a Logistic Regression classifier to estimate churn probability per subscriber, selected for interpretability of its coefficients alongside strong predictive performance (ROC-AUC 0.88, recall 0.84).
- Segmentation: applied K-Means clustering to the 2,874 subscribers flagged as medium- or high-risk, grouping them into three behaviorally distinct segments based on engagement level, plan tier, and login recency.
- Decision framework: mapped each segment to a specific retention action and quantified expected impact using a stated, adjustable assumption set (below), rather than a single blanket recommendation.
- Delivery: published an interactive Tableau dashboard presenting the risk drivers, segment structure, and recommended actions for stakeholder review.

Key Findings

- Engagement is the dominant churn driver: subscribers watching under 5 hours churn at 72.9%, versus 23.4% for those watching 15+ hours.
- Plan tier compounds risk independently of engagement: the Basic plan churns at 61.8%, roughly 1.4x the rate of Standard or Premium tiers.
- Login recency and watch time are the two strongest predictors in the model, outweighing demographic factors (age, gender, region) which showed no meaningful relationship to churn.
- At-risk subscribers are not a single population: three distinct segments emerge, each requiring a different intervention rather than one uniform offer.

Segments, Root Causes & Recommended Actions

Segment	Size	Root cause	Recommended action	Est. value at risk recovered
Higher-value, going quiet	907	Pays above-average fees; median 50 days since last login	Proactive win-back outreach: one month free or a complimentary plan upgrade	\$34,366
Basic plan, going quiet	1,185	Lowest-cost plan; median 37 days since last login	Low-cost reactivation: automated reminder email/push with a free-week offer	\$25,568
Barely watching	782	Engagement has collapsed to under 3 hours watched	Content-led re-engagement: personalized genre-based recommendation email	\$25,392

Impact estimates are based on a stated assumption set: a targeted intervention recovers 25% of subscribers otherwise expected to churn within a segment, and retained-subscriber value is calculated as 12 months of that subscriber's monthly fee. This assumption is intentionally explicit so it can be adjusted against actual historical win-back rates.

Tools & Data

Python (pandas for data preparation; scikit-learn for classification and clustering) for the analytical pipeline. Tableau for the stakeholder-facing dashboard. Dataset: 5,000-subscriber streaming platform sample (subscription tier, watch hours, login recency, monthly fee, churn outcome).